

Fig. 1.10

$$(-7)+\left[(-2)+(-9)\right] = (-18)$$

We reach the same number –18 in both the cases. Hence, regrouping of integers does not change the value of the sum. This property is known as "associative property" under addition.

Therefore, for any three integers a, b, c; $a+(b+c) = (a+b)+c$

The collection of integers has positive, negative integers and zero. Have you noticed that zero is neither positive nor negative integer. What happens when we add zero to an integer?

For example, we observe that 7 + 0 = 7, –3 + 0 = –3, –27 + 0 = –27,
–79 + 0 = –79, 0 + (–69) = –69, 0 + (–85) = –85.

From the above it is clear that whenever zero is added to an integer, we get the same integer. Due to this property, zero is called the identity with respect to addition or "additive identity" of the collection of integers.

Therefore, for any integers a, $a + 0 = a = 0 + a$

The additive identity zero separates the number line into positive and negative integers. We have +1 and–1,+5 and –5,–15 and +15, etc. on opposite sides of the number line that are equidistant from zero. Such integers on either side of zero are called "opposites" of each other. In fact, we find that the "opposites" added together always give the value zero.

For example, $(-15)+15=0$, $21+(-21)=0$. This property of integers is named as "additive inverse". (–15) is the additive inverse of 15 because their sum is zero. In the same way, 21 is the additive inverse of -21. Either of the pair of opposites is known as the "additive inverse" of the other.

Therefore, for any integer a, $-a$ is the additive inverse.

$a+(-a) = 0 = (-a)+a$

Try these

1. Fill in the blanks:
   (i) $20+(-11)=___+20$ (ii) $(-5)+(-8)=(-8)+___$ (iii) $(-3)+12=___+(-3)$
2. Say true or false.
   (i) $(-11)+(-8) = (-8)+(-11)$ (ii) $-7+2=2+(-7)$ (iii) $(-33)+8=8+(-33)$

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